

Geothermal Well Testing

NESR

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NESR Experiences in Geothermal field

NESR in Indonesia:

PT NPS Energy Indonesia

PT Tiger Energy Services ROI

Well Stimulation	Perforation	Capillary String	Production Chemical









NESR Experiences in Geothermal field

NPS Energy

tiger

Well Testing



Wireline Logging



Wireline Mast



WHAT ?

WHY ?

Geothermal Well Testing

Geothermal field development: the sequence:

1. Exploration.
   Exploration wells: Drilling & Well Testing.
2. Development
   Infrastructures construction
   Drilling production & injection wells.
   Well Testing
   Pipeline construction
   Power plant construction
3. Power generation

Relative cost of Well Testing per well*:

1. Value of geothermal reserve: 0 – 4000
2. Drilling Cost: 100
3. Well Testing: 1

*Note: MB Century, NZ, 2010

Geothermal Well Testing

The purpose is to obtain information:

1. Individual well characteristics.
2. Location of permeable zones.
3. Reservoir permeability characteristics.
4. For reservoir development.
5. Well discharge characteristics: flowrate (total & each phase), enthalpy, fluid chemistry.
6. Sub-surface temperature & pressure distribution.

Geothermal Well Testing

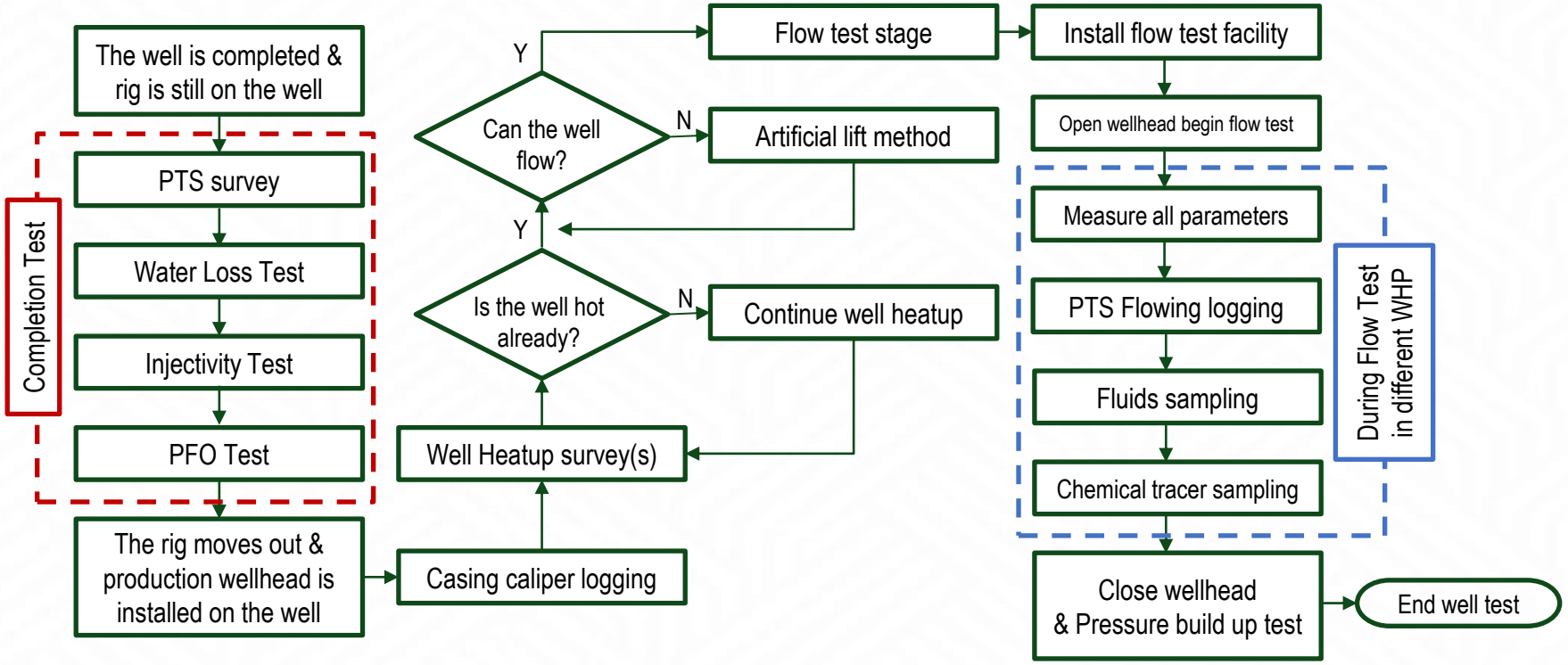
Well Testing is therefore undertaken during following periods:

1. During drilling,
2. At the completion of drilling operations,
3. Initial shut-in period,
4. Flow testing,
5. Before well starts commercial operation.

Once on commercial operation, ability to test the well becomes more difficult to schedule and implement.



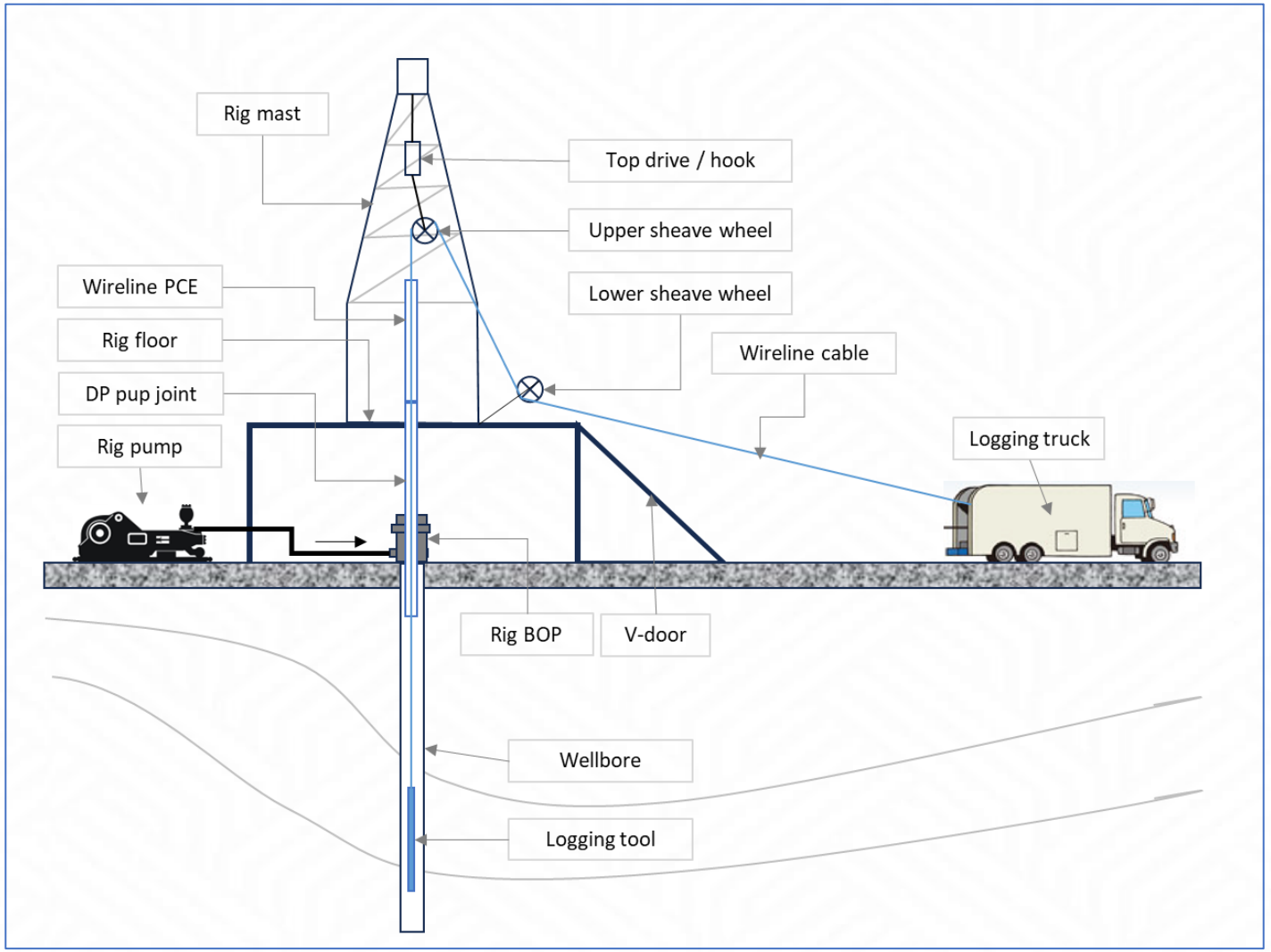
Typical Geothermal Well Test Flowchart, for new well



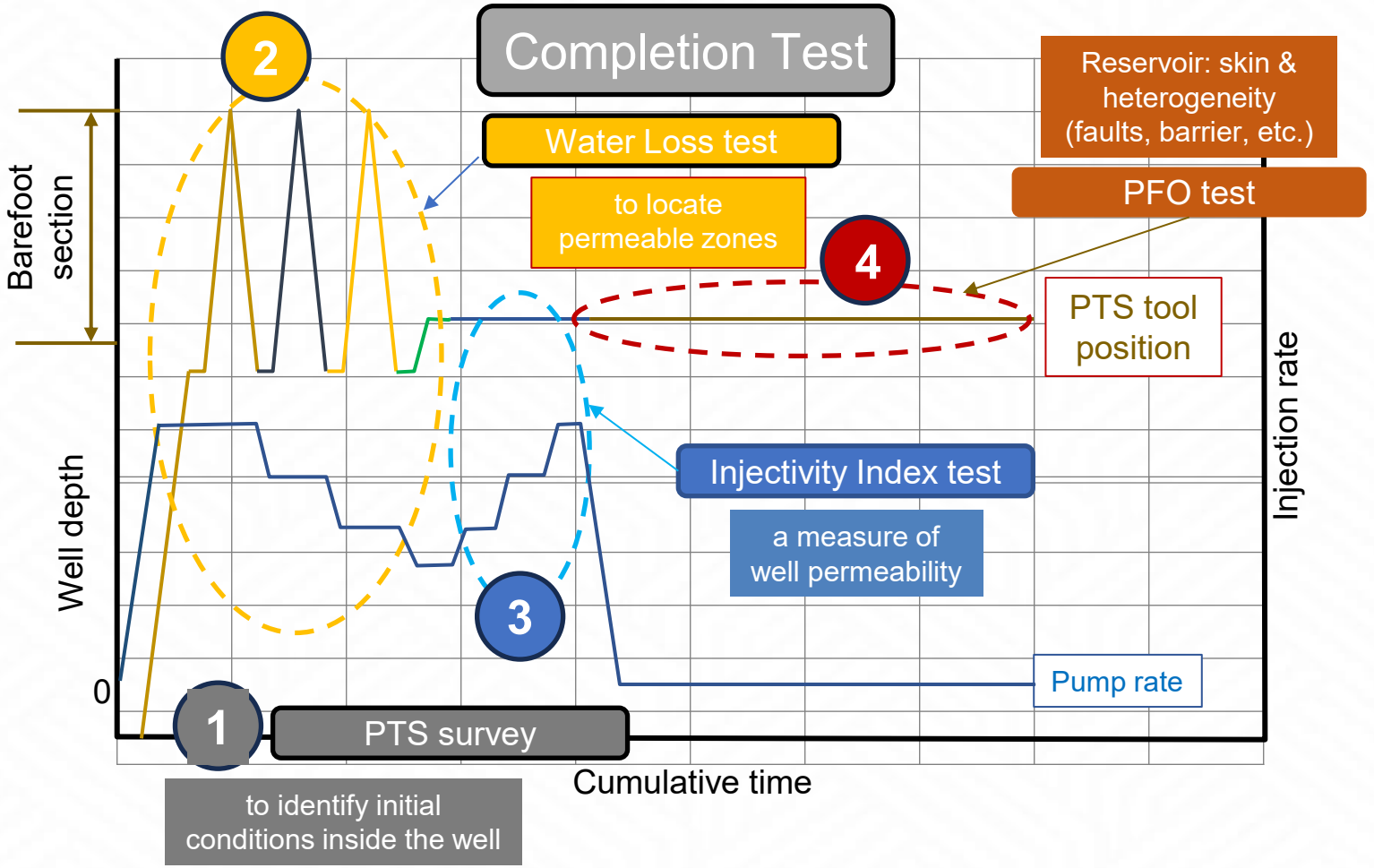
Geothermal Well Tests:

1. **Completion Test**: Wireline Logging & Pumping
2. **Well Integrity Test**: Wireline Logging
3. **PT Heat-Up Test**: Wireline Logging
4. **Flow Test**: Short-term & Long-term Test
5. **PTS Flowing**: Wireline Logging during Flow Test
6. **Geochemistry Sampling**: during Flow Test
7. **Chemical Tracer Test** (aka TFT®): to validate the Flow Test results.

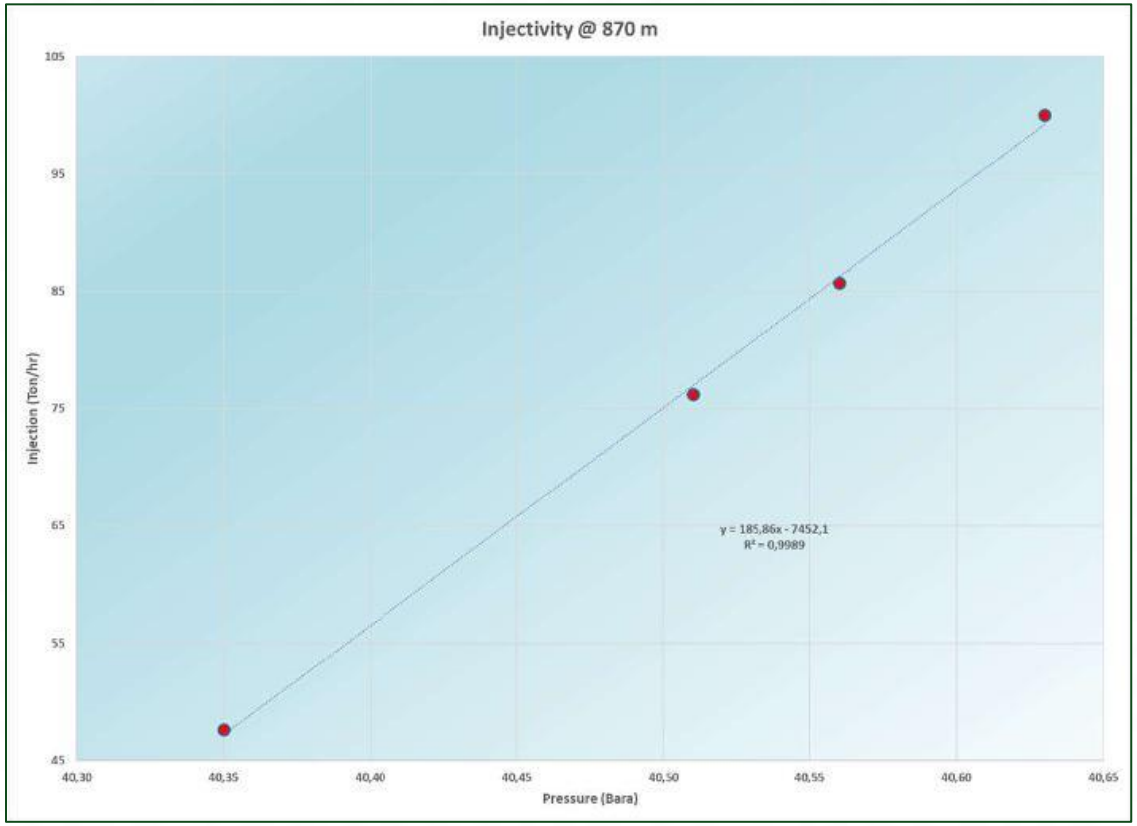
Geothermal **Completion Test** – typical equipment setup with the Rig still in operation.



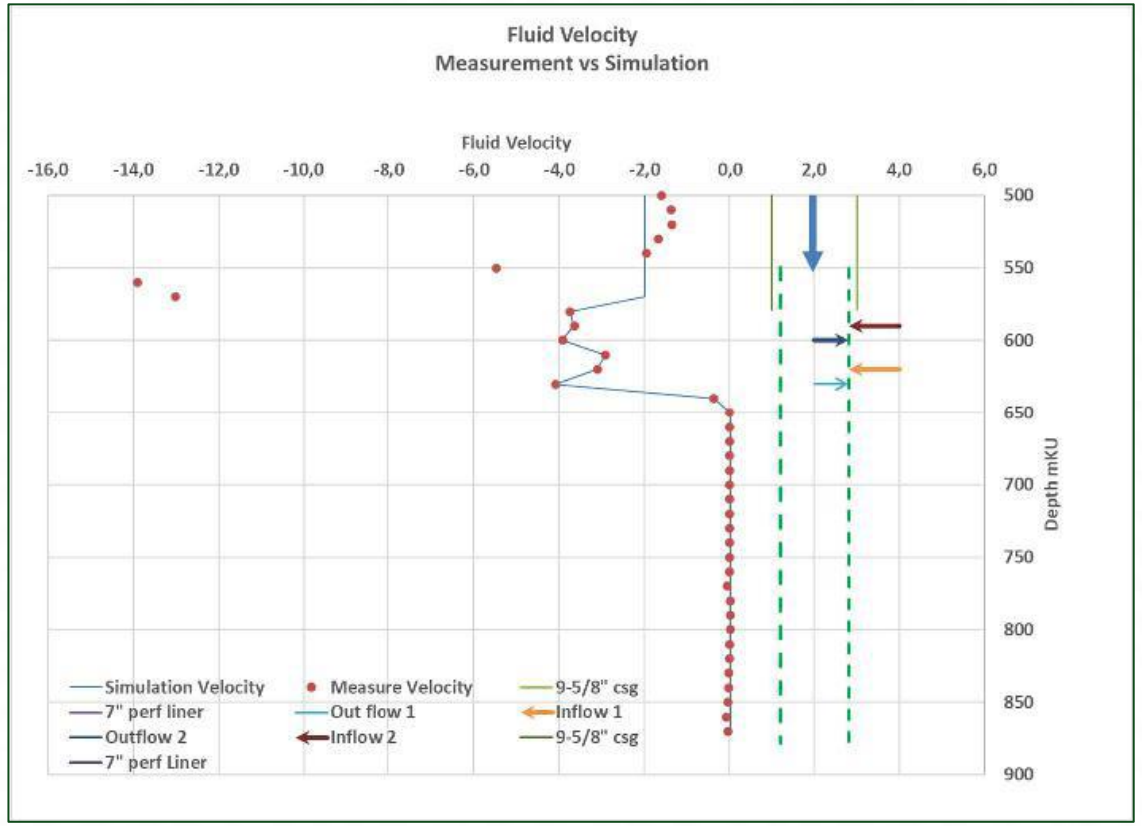
Typical Geothermal **Completion Test** program



PTS Injection, example & analysis

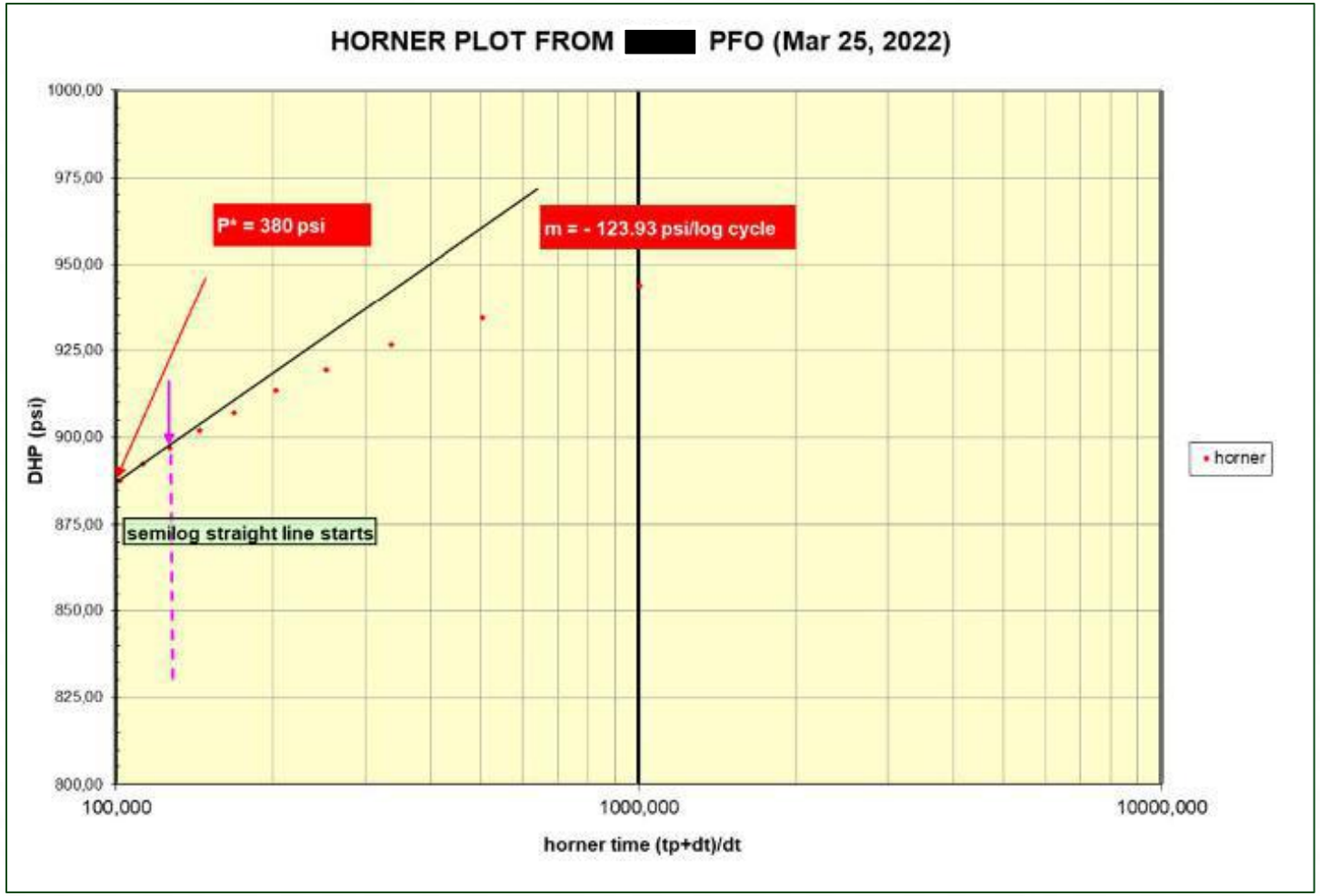


Injectivity at 870m depth.
Injectivity: 185.86 ton/hr/bar



Fluid velocity plot: Measurement vs Simulation along the production zone

PFO (Pressure Fall Off) test: example & analysis



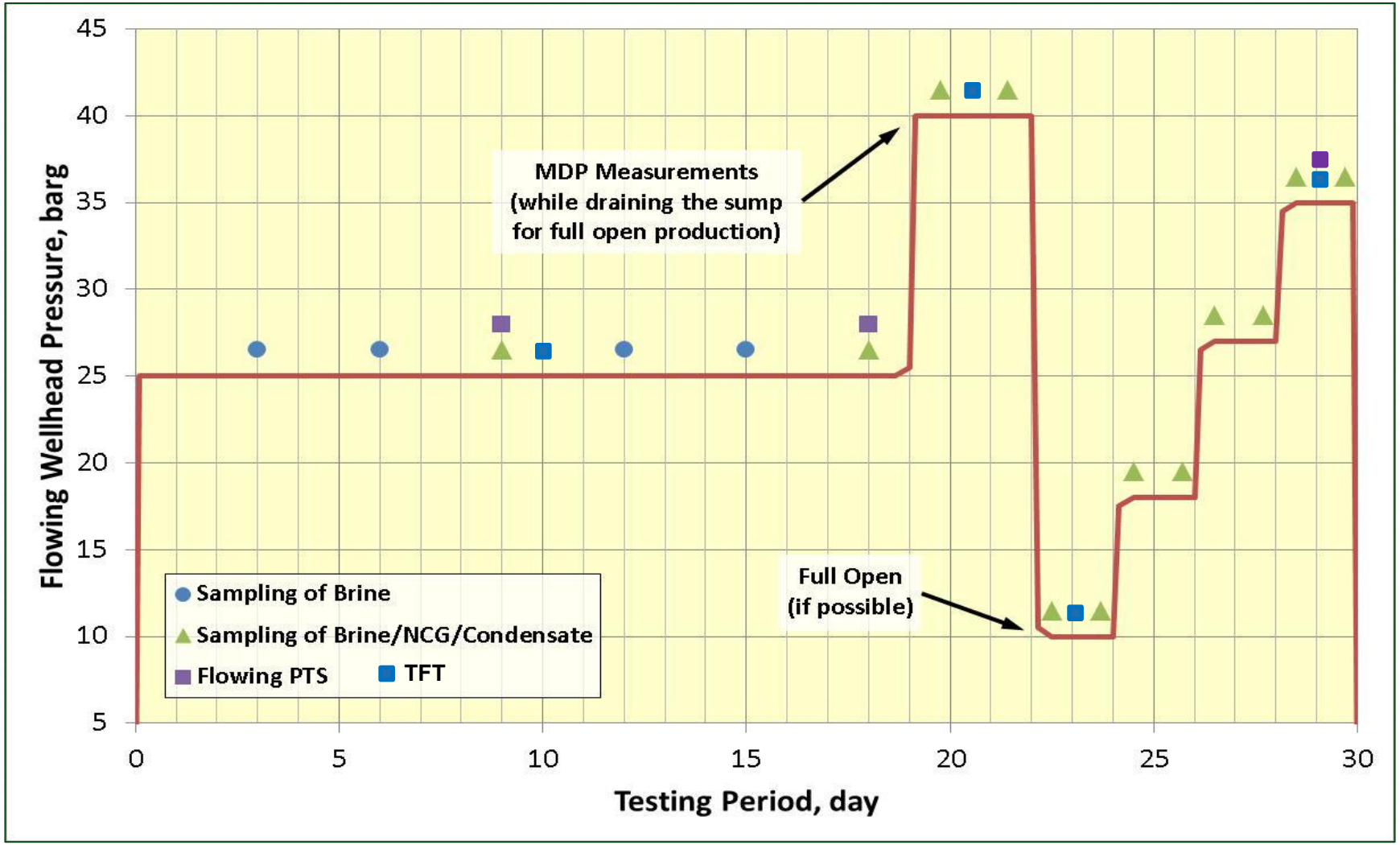
Calculated:
Permeability thickness (kh) : 3,057md-ft
Permeable zone thickness (h) : 1,290ft
Permeability (k) : 2md (→ low)
Skin (s) : -11,4 (→ excellent)



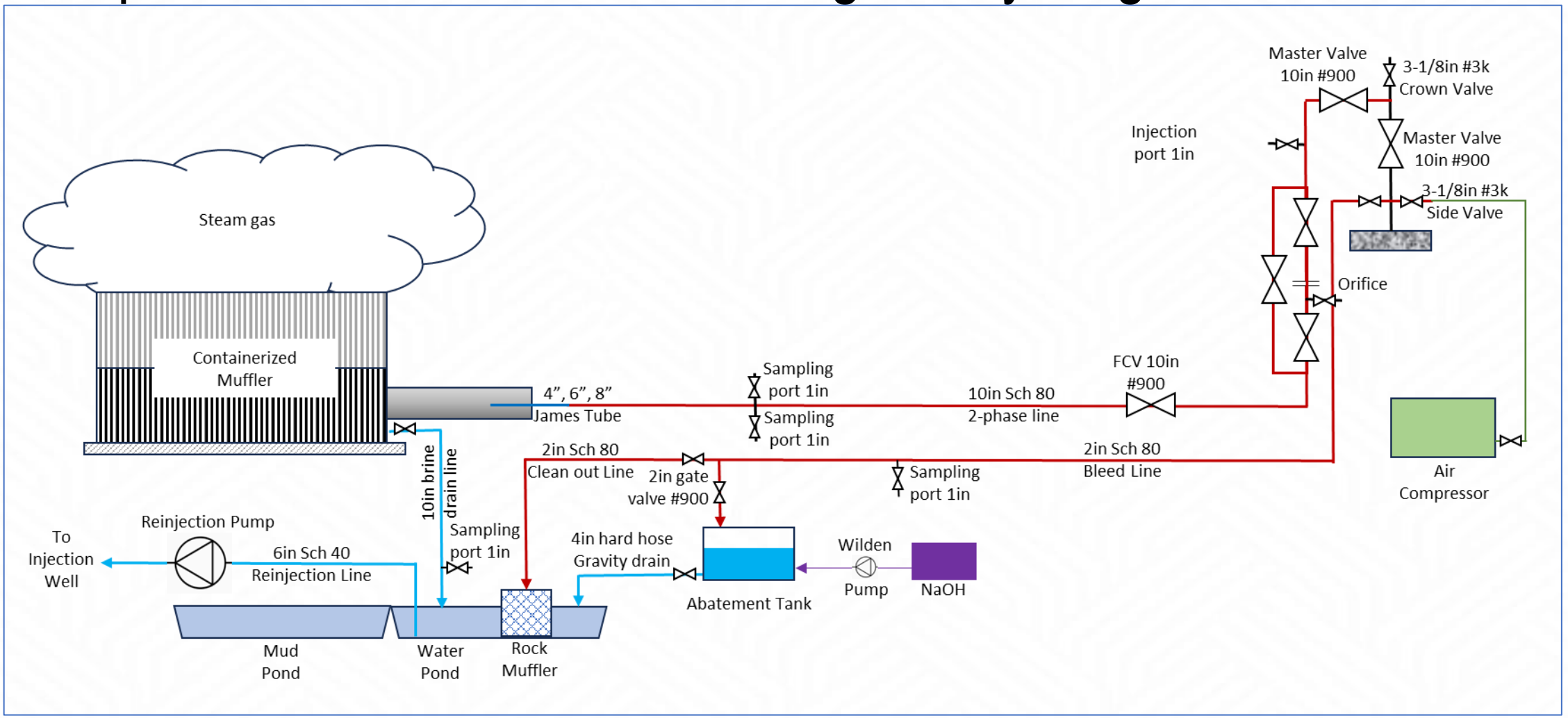
Geothermal (Long Term) Flow Test

1. Determine **well discharge characteristics:**
   - changes with wellhead pressure, changes with time,
   - **enthalpy** and **mass flow**,
   - **fluid chemistry**.
2. Determine **plant design** and **operating parameters:**
   - **pipeline** sizing,
   - optimum **separation/inlet** pressures,
   - steam/gas **processing capacity**,
   - **corrosion/scaling mitigation** requirements.
3. Determine **production** and **injection well requirements.**
4. Provide information on **réservoir conditions.**

Typical Geothermal Flow Test program



Complete Geothermal Flow Testing facility diagram



Geothermal Well Testing

Flow Test Separator / Muffler

→ To separate the two-phase well fluid between gas phase (steam) & liquid phase (brine).





Containerized Muffler, the advantages

1. Easy to transport, mob-demob & setup.
   - Constructed using a standard ISO container.
   - Available in both 40ft & 20ft containers.
2. High-efficiency separation of brine.
   - 1st separation: Perforated pipes
   - 2nd separation: Mist pads on top of container.
3. Lower noise emissions than standard muffler.
   - Double wall.
   - Easy to add sound barrier.
4. Greater capacity.
   - Max WH mass flowrate: 870T/hr (40ft) & 435T/hr (20ft).
   - Equivalent to a 5.4m diameter single stack or 3.8m diameter twin stack (for 40ft container size).
5. Better accuracy.
6. Lower cost compared to cyclone mufflers & production separators.

Geothermal Well Testing – Flow Test

Methods to obtain mass flowrate during Flow Test:

1. **Russel James method** (Lips pressure): main & most popular.
2. **Orifice**: redundant.
3. **Chemical Tracer Test** or TFT(Tracer Flow Test)^{TM} : validation.

Geothermal Flow Test – Russel James method

Russel James method (Lips pressure), requires:

1. Lip pressure, P_c: measured
2. Lip pipe internal area, A: measured
3. Brine flow rate (atmospheric condition), W_{atm}

- Weir box: measured & calculated (complicated & less accurate).
- Digital flowmeter: measured (easy, fast & accurate).

Steps:

1. Calculate the enthalpy.
2. Calculate Total Mass Fluid Rate.

Geothermal Flow Test – Russel James method

Parameters	Symbols	Data sources
Lip Pressure	P_c	Pressure transmitter/gauge in Lip Pipe $P_{c,abs} = P_{c,\ gauge} + P_{atm}$
Inside diameter lip pipe	D_i	Actual dimension of internal pipe
Head above V-notch	H_w	Weir box
Brine temperature in weir box	T_w	Temperature gauge
Weir box: width	$B,\ D$	Fabrication dimensions
Local atmospheric pressure	P_{atm}	Barometer

Geothermal Flow Test – Russel James method

Steps:

1. Internal Lip Pipe area $$A = \frac{\pi D_i^2}{4}$$

2. Brine flowrate (Weir box method)

$$K = 81.2 + \frac{0.24}{H_w} + \left(\frac{12}{\sqrt{D}} + 8.4\right)\left(\frac{H_w}{B} - 0.09\right)^2$$

then: $$W_{\text{atm}} = K\, H_w^{2.5}\, \rho_w\, (0.06)$$

3. James factor calculation $$Y = \frac{W_{\text{atm}}}{A\, P_c^{0.96}}$$

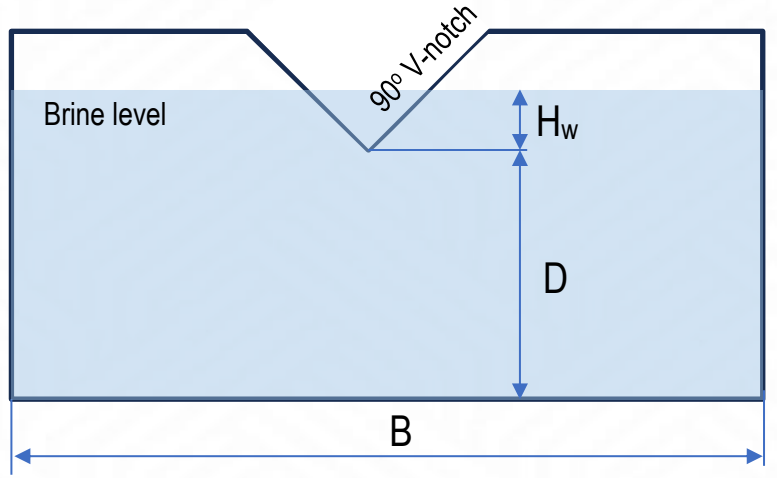
4. Flowing enthalpy $$h = \frac{h_{g,\text{atm}} + 925Y}{1 + 7.85Y}$$

h = well total flowing enthalpy (kJ/kg)
$h_{g,atm}$ = saturated steam enthalpy on the actual atmospheric pressure (kJ/kg) → from steam table
Y = James factor

5. **Total mass flowrate** $$\dot{m}_t = \left(\frac{0.184}{h^{1.102}}\right) A\, P_c^{0.96}\, (3600)$$

m_t = total mass flowrate (Ton/hour)

W_{atm} = brine flowrate in weir box (ton/hour)
H_w = brine level above the weir vertex (meter)
this is what we measure regularly on the field
D = weir vertex to weir box floor (meter)
B = weir box width (meter)
ρ_w = brine density on a specific temperature (kg/m^3)
K = correlation discharge coefficient



Geothermal Flow Test – Orifice Plate method

For single-phase flow, the total mass flow rate is calculated as:

$$m = \frac{C}{\sqrt{1-\beta^4}} \cdot \epsilon \cdot \frac{\pi}{4} \cdot d^2 \cdot \sqrt{2 \cdot \rho \cdot \Delta P}$$

or simplified into a Meter Constant formula:

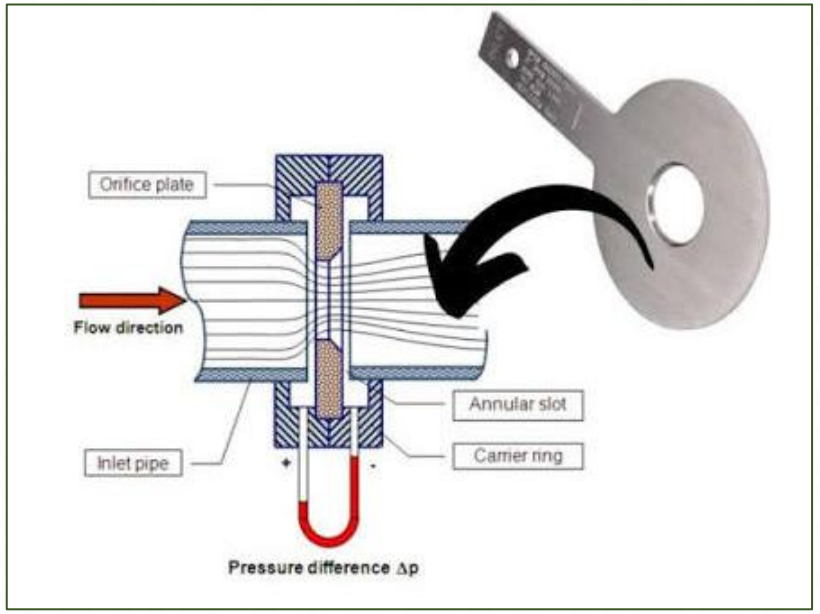
$$m = K \cdot \sqrt{\rho \cdot \Delta P}$$

Steps:

1. Collect the measured data & constraints:
   a. Differential Pressure from the orifice plate pressure transmitter: ΔP.
   b. Line Temperature LT or Pressure LP: needed to find the fluid density ρ from steam tables.
   c. Orifice diameter d.
   d. Pipe diameter D.
2. Determine fluid density ρ & fluid phase.
3. Calculate the Beta Ratio β

$$\beta = \frac{d}{D}$$

4. Discharge Coefficient C: typical = 0.60 – 0.62.
5. Expansibility Factor ϵ:
   a. Single-phase Liquid (Brine) ϵ = 1.0.
   b. Single-phase Dry Steam: use ISO 5167 formula. Typical ϵ ~ 1.3.



Geothermal Flow Test – Orifice Plate method

For two-phase flow, the total mass flow rate is calculated

$$m_t = \frac{C \cdot \epsilon \cdot \frac{\pi}{4} \cdot d^2 \cdot \sqrt{2 \cdot \rho_{app} \cdot \Delta P}}{\sqrt{1-\beta^4}}$$

ρ_{app} *:* apparent two-phase density

$$\frac{1}{\rho_{app}} = \frac{x^{1.5}}{\rho_g} + \frac{1-x^{1.5}}{\rho_f}$$

x : Steam dryness fraction (mass of steam / total mass).

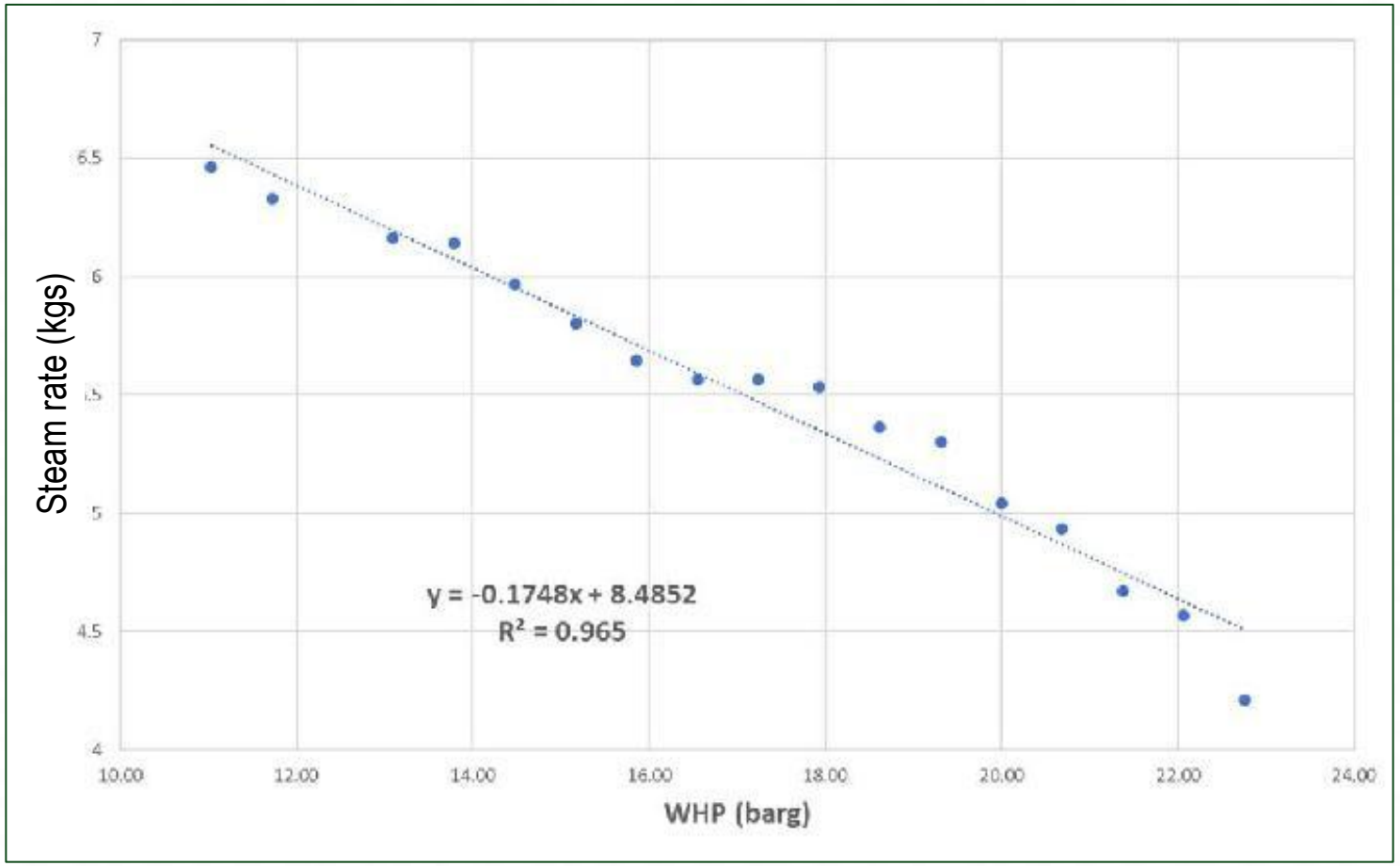
ρ_g : Density of pure saturated steam (kg/m^3)

ρ_f : Density of pure saturated water (kg/m^3)

Steps:

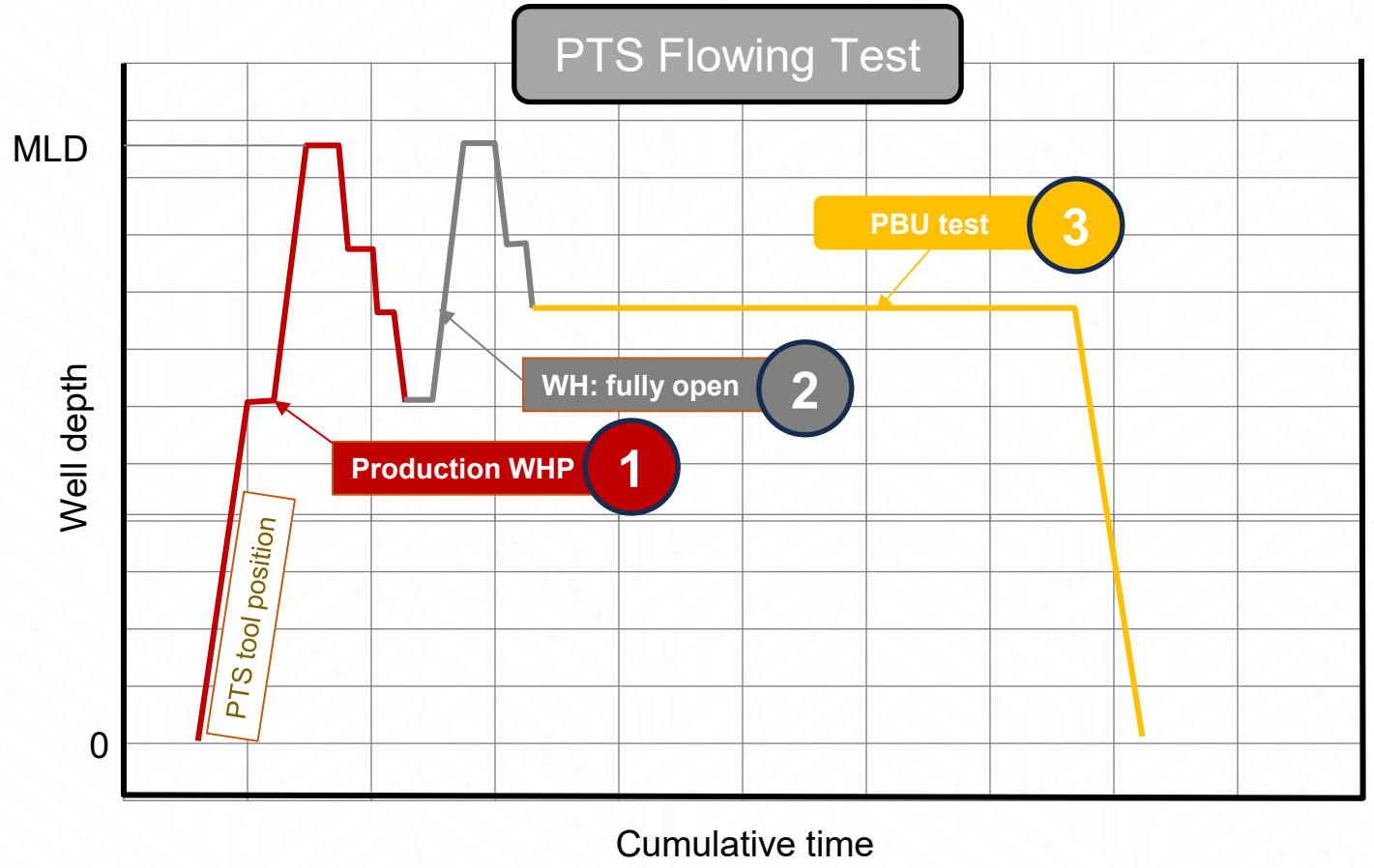
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   a. Differential Pressure from the orifice plate pressure transmitter: ΔP.
   b. Line Temperature LT or Pressure LP: needed to find the fluid density ρ from steam tables.
   c. Orifice diameter d.
   d. Pipe diameter D.
2. Determine fluid density ρ & fluid phase.
3. Calculate the Beta Ratio β $\beta = \frac{d}{D}$
4. Total Fluid Enthalpy (H). must be known beforehand from recent tracer dilution tests, a separation test, or historical well performance data (kJ/kg).
5. Use LP & saturated steam tables to get:
   ρ_f (Liquid Density) & ρ_g (Steam Density).
   h_f (Liquid Enthalpy) & h_g (Steam, Enthalpy).
6. Calculate Steam Dryness Fraction (x): $x = \frac{H-h_f}{h_g-h_f}$
7. Calculate Apparent Two-Phase Density ρ_{app}
8. Solve for Total Mass Flow Rate
   use a standard two-phase discharge coefficient
   C ≈ 0.61 & ϵ ≈ 1.0.

Steam Deliverability Curve from flow test: example

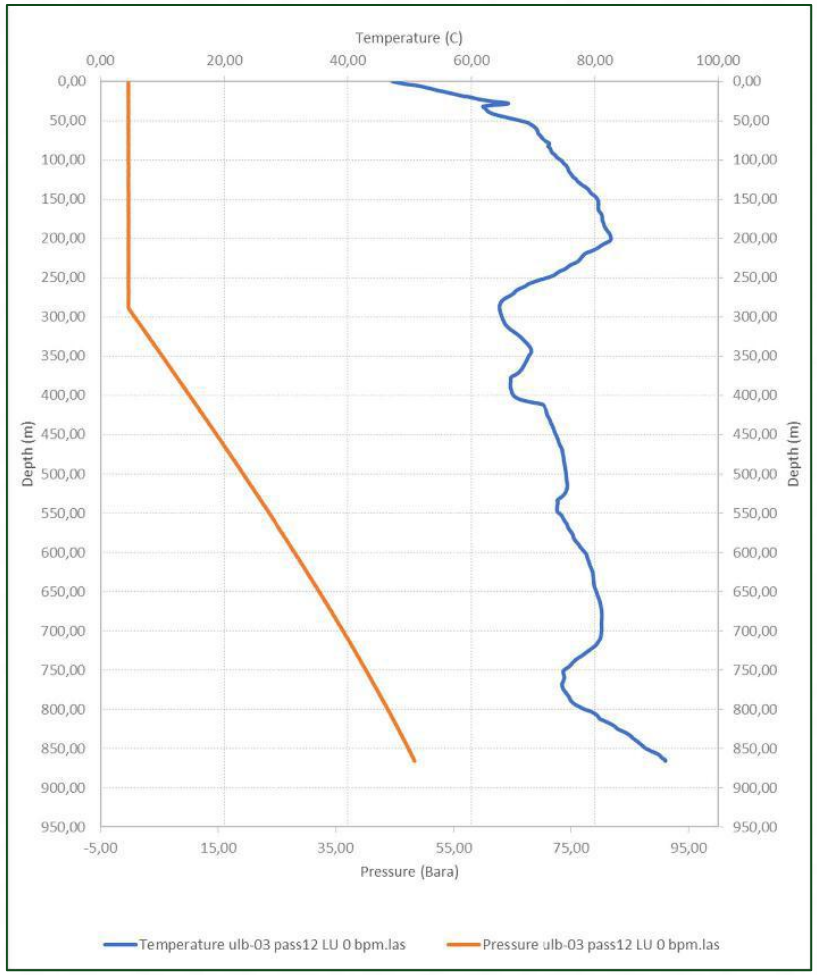


Steam **deliverability curve** along various WHP

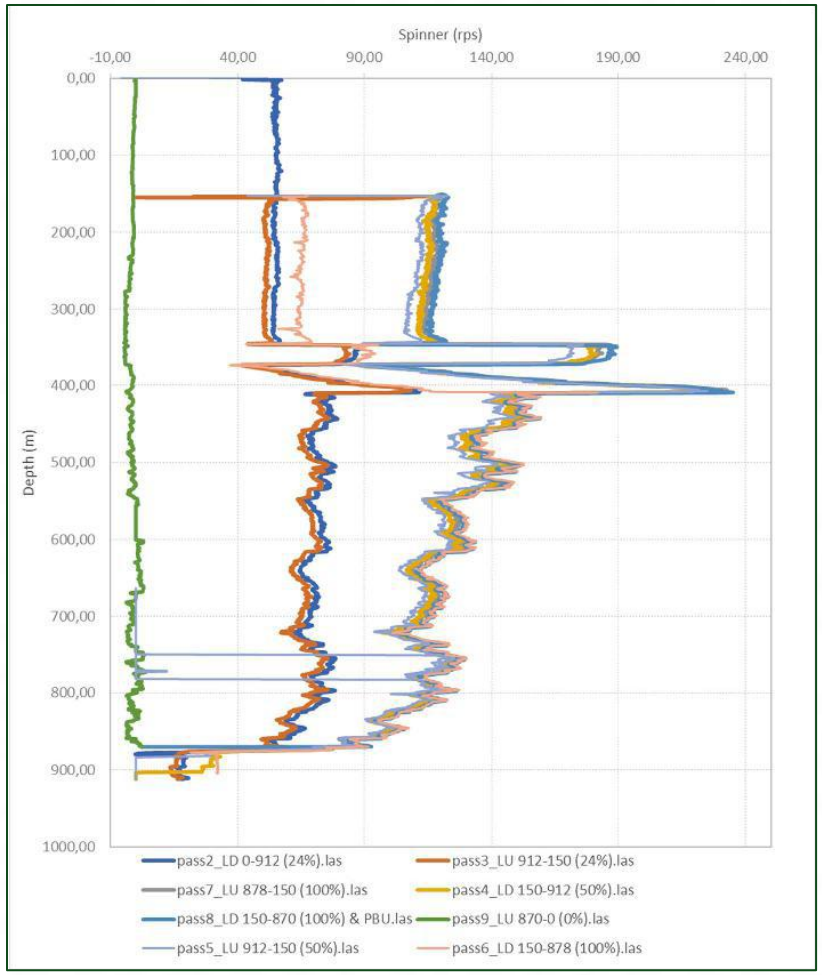
Typical Geothermal PTS Flowing Test program



PTS (Pressure Temperature Spinner) sample graphs

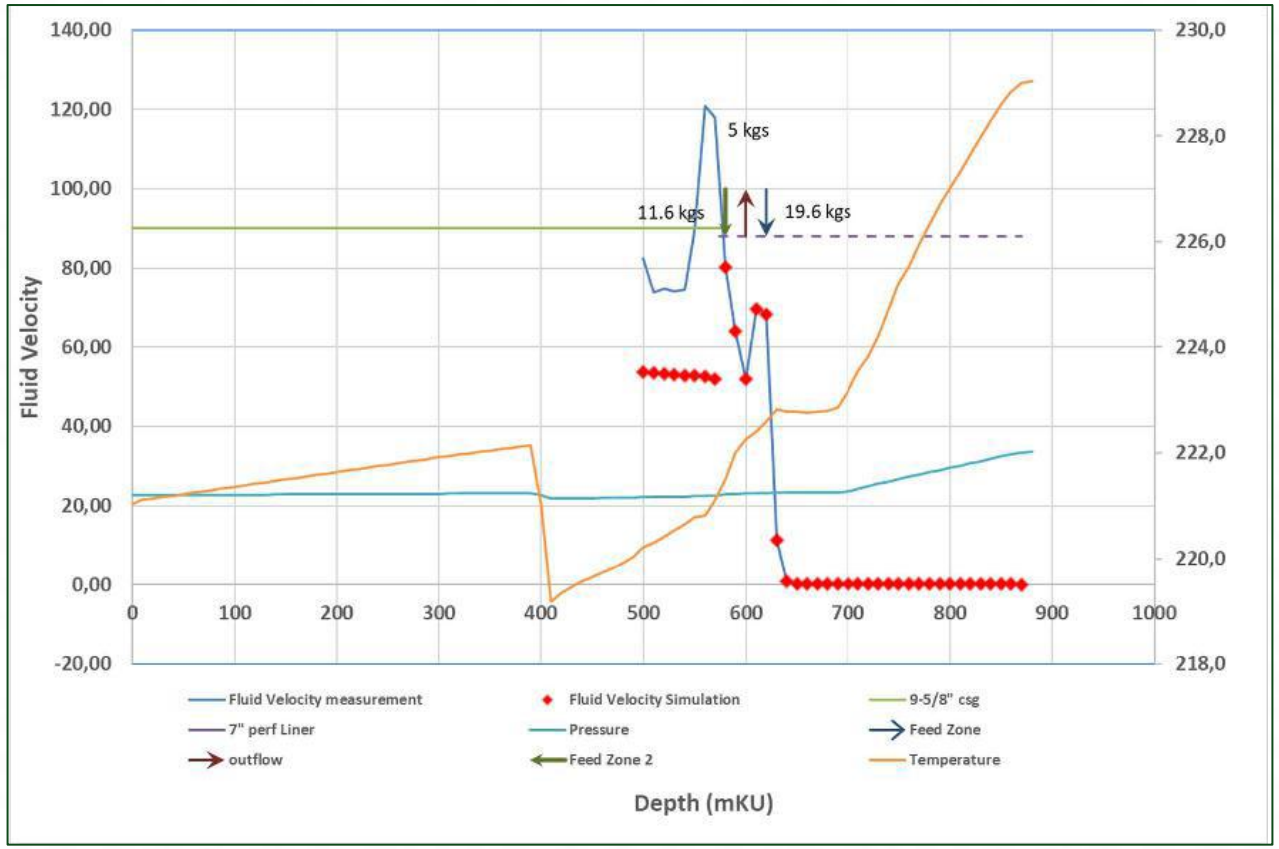


Pressure and Temperature vs Depth chart



Spinner vs Depth chart

PTS Flowing, example & analysis



Fluid velocity plot: Measurement vs Simulation, along the 300m production zone

Analysis:

1. Along 300m of zone of interest, there are only 90m of inflow, 10m of outflow and 200m long non-production zone.
2. The total pay zones produce 26.38kg/s of steam, including 5kg/s lost due to a thief zone / low pressure zone (outflow).
3. The largest pay zone, 16.24kg/s is produced at 620m depth.
4. The thief zone presents at 600m depth.
5. The non-production zone is at depth of 650-850m.

Geochemistry Sampling: example

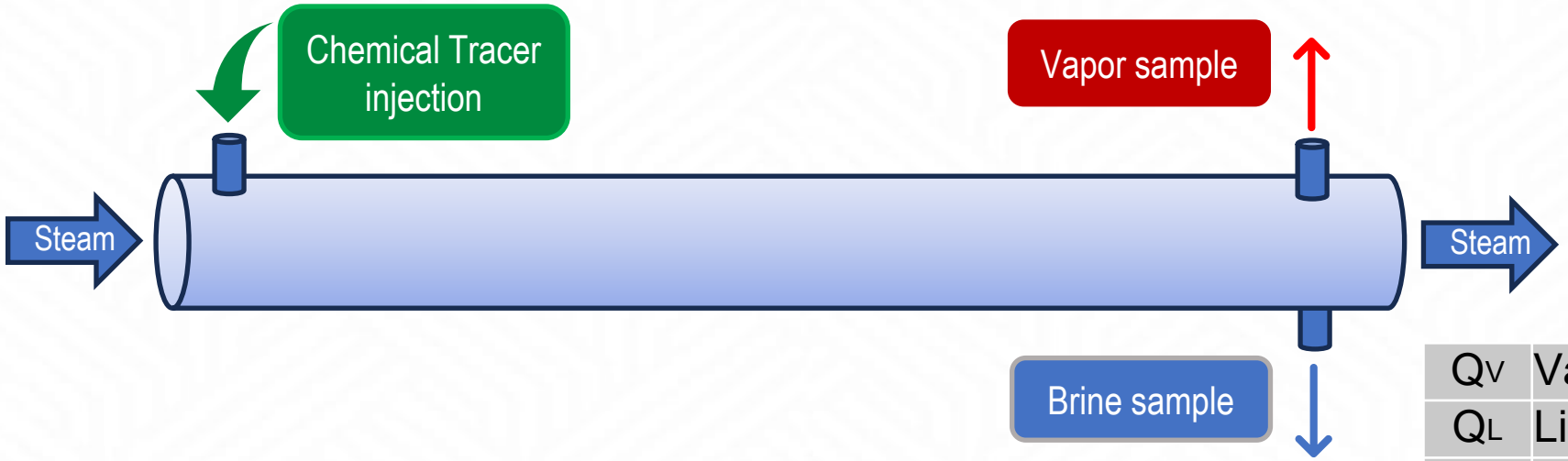
No	Date	Sample Label		Sodium (Na)	Chloride (Cl-)	Boron (B)	Silica (SiO2)	Arsenic (As)	Mercury (Hg)	Iron (Fe)	pH	Bicarbonate ($HCO3^-$)	Sulphate ($SO2^-$)4
			Det. Limit Unit	0.005 mg/L	0.005 mg/L	0.02 mg/L	0.02 mg/L	0.0005 mg/L	0.0005 mg/L	0.03 mg/L		1 mg/L	0.2 mg/L
1	April 7, 2022	3/07.04.22/09: 32/WITA/SC		4.57	2.19	1.21	0.10	0.0092	<0.0005	<0.03	6.79	132	1.4
2	April 8, 2022	3/08.04.22/17: 43/WITA/SC		0.383	0.09	<0.02	<0.02	0.0024	<0.0006	0.04	6.78	461	4.1
3	April 10, 2022	3/10.04.22/09:27/WITA/SC		0.117	0.019	<0.02	<0.02	0.0019	<0.0007	<0.03	7.02	232	4
4	April 14, 2022	3/14.04.22/08:45/WITA/SC		0.067	0.387	0.91	<0,02	0.0022	<0.0008	0.1	6.21	64	3.2

Table Condensate analysis

No	Date	Sample Label	Carbon Dioxide (CO2)	Hydrogen Sulfide (H2S)	Ammonia (NH3)	Argon (Ar)	Nitrogen (N2)	Methane (CH4)	Hydrogen (H2)	NCG	Air
			% by Volume	% by Volume	% by Volume	% by Volume	% by Volume	% by Volume	% by Volume	% wt	%
1	April 7, 2022	07.04.22/09:22/WITA (W99)	95.14	0.0061	0.152	0.035	3.924	0.0166	0.728	4.10	0.026
2	April 8, 2022	08.04.22/17:15/WITA (W105)	92.31	0.0341	0.049	0.055	6.125	0.0187	1.405	26.92	0.017
3	April 10, 2022	10.04.22/09:19/WITA (W67)	93.25	0.0194	0.042	0.042	5.456	0.0194	1.175	20.10	0.597
4	April 14, 2022	14.04.22/08:45/WITA (W7)	95.08	0.0125	0.087	0.038	4.283	0.0123	0.489	4.14	0.563

Table NCG analysis

Chemical tracers / Tracer Flow Test ™



Mass flowrate:

Liquid: $Q_L = Q_{TL} / (C_L - C_B)$

Vapor: $Q_v = Q_{TV} / (C_v - C_B)$

- Enthalpy and flow rate determined by measuring dilution of vapor and liquid tracers injected into two-phase line.
- Separated steam and water collected downstream of injection point.
- Water and steam samples analyzed for tracer content.

Q_V	Vapor Mass Flowrate
Q_L	Liquid Mass Flowrate
Q_{TV}	Vapor Tracer Mass Injection Rate
Q_{TL}	Liquid Tracer Injection Rate
C_V	Tracer Concentration in Vapor
C_L	Tracer Concentration in Liquid
C_B	Background Concentration of Tracer in Respective Phase

Calculating the Well's Electric Capacity.

Specific Steam Consumption (SSC):

- A standardized industry conversion factor.
- It represents how many kilograms of steam per second are required to generate 1 Megawatt of electricity (kg/s per MWe).
- Values: 1.8 – 2.5 kg/s per MWe (for modern geothermal condensing turbines, depending on turbine efficiency & separation pressure).

For a Single-Phase Dry Steam Well

The formula:

$$\text{Capacity (MWe)} = \frac{\text{Total Steam Mass Flow Rate } (m_g) \text{ [kg/s]}}{\text{Specific Steam Consumption (SSC) [kg/s/MWe]}}$$

For a Two-Phase Well (Flash Steam Plant):

- Calculate Separated Steam Flow (m_g)

$$m_g = m_t \cdot x$$

- Convert Steam Flow to MWe

$$\text{Capacity (MWe)} = \frac{m_g}{\text{SSC}}$$

The Thermodynamic Exergy Method (Advanced)

- The formula:

$$\text{Capacity (MWe)} = m_g \cdot (h_{in} - h_{out}) \cdot \eta_t \cdot \eta_g \cdot 10^{-3}$$

- h_{in}: Enthalpy of the steam entering the turbine (kJ/kg)
- h_{out} : Enthalpy of the steam exiting the turbine / condenser (kJ/kg).
- η_t: Turbine polytropic efficiency (typically 0.75 - 0.85).
- η_g: Generator efficiency (typically 0.95 - 0.98).

Geothermal Well Testing – Q & A



Ulumbu, Flores, Apr 2022